CARBON QUANTUM DOTS OVERVIEW

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1. Introduction

The unique properties of carbon-based nanomaterials such as carbon nanotubes, graphene sheets and fluorescent carbon nanoparticles, or carbon quantum dots (CQD) have motivated wide-ranging studies due to their great potential for a wide variety of technical applications. Among the electronic and physical chemistry qualities of CQDs, their optical properties and fluorescence emissions have increased the amount of research being performed on them, especially in recent years. In addition, CQDs possess the intrinsic advantages of being an abundant element and having relative low toxicity. The latter being manipulated in the use of CQDs as fluorescent tags for imaging applications.

The general description of CQDs is that of nanometer size particles consisting of sp2 hybridized graphitic core functionalized with polar carboxyl or hydroxyl groups on the surface. Initial CQD synthesis efforts focused on top-down methods such as laser ablation or electrochemical oxidation [1].

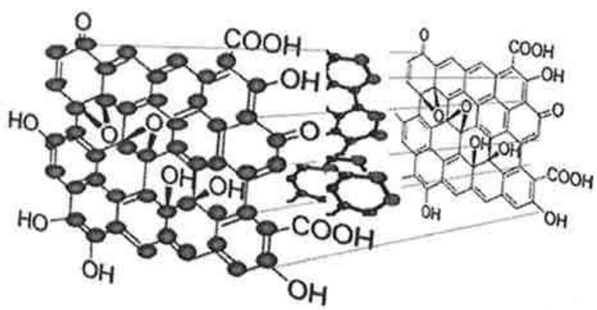

Figure 1: Chemical structure of CQDs [2]

As a newly emerged type of specific fluorescent nanomaterials, CQD's have shown tremendous potential as versatile nanomaterials for a wide range of applications: drug delivery, chemical sensing, photodynamic therapy, photocatalysis and electrocatalysis. When being compared to contemporary or conventional semiconductor quantum dots, the attributes of CQDs' simplistic functionalization and good physical-chemical and photochemical stability makes them appealing for technical applications. Together with other advantages such as low cost and ease of synthesis [2], CQDs are in a favorable position for achieving unprecedented performance.

2. Synthesis, size control and modification

During the last decade, several development procedures have been developed to prepare and improve CQDs, which can be loosely divided into "Top-down" and "Bottom-up" solutions. Both can be changed during a pretreatment, prep phase or a post-treatment phase. Three problems in the preparation of CQDs need to be noted:

(a) carbonaceous accumulation during carbonization, which can be prevented utilizing electrochemical synthesis, continuous pyrolysis or solution chemistry methods,

(b) size regulation and uniformity, which is critical for uniform properties and mechanistic analysis, and can be improved by post-treatment, such as gel electrophoresis, centrifugation and

(c) surface properties that are critical for solubility and selected applications [5].

2.1 Chemical Ablation

Strong oxidizing acids carbonize small organic molecules to carbon-rich materials. These materials can be further cut into small sheets through controlled oxidation. However, this method is associated with harsh conditions and severe processes. The emission wavelength of these CQDs can be modified by differing the starting material and the length of a nitric acid treatment [5]. The multicolor emission capabilities and nontoxic nature of the CQDs produced allow them to be applied in life science research.

2.2 Electrochemical Carbonization

The preparation of CQDs from electrochemical soaking is a powerful method in production. The process uses various carbon materials in bulk as precursors. The electrochemical method involves a top-down solution, where the chemical cutting process of a carbon material, such as carbon nanotubes or graphite, is used [5]. However, carbon dots prepared through contemporary electrochemical methods have varying sizes and would require further separation through filtration or chromatography to obtain uniformity.

2.3 Microwave Irradiation

The use of microwave irradiation is a low cost and rapid method in the synthesis of CQDs. Sucrose is used as the carbon source with diethylene glycol (DEG) as media for reaction. Through one minute of microwave radiation, green luminescent CQDs were obtained. The resultant CQDS are highly bio-compatible and have increased potential for applications in the biomedical field.

2.4 Hydrothermal/solvothermal Treatment

Hydrothermal carbonization (HTC) or solvothermal carbonization is a low cost, environmentally friendly and non-toxic route for the manufacture of innovative carbon-based products from specific precursors. CQDs were produced with sizes of 1.5–4.5 nm and were used in bioimaging because of their high photostability and low toxicity [3]. Solvothermal carbonization followed by extraction with an organic solvent is a popular approach to prepare CQDs. Typically, carbon-yielding compounds were subjected to heat treatment in high boiling point organic solvents, followed by extraction and concentration.

Synthetic Methods	Advantages	Disadvantages	Ref.
Chemical ablation	Most accessible, various sources	Harsh conditions, drastic processes, multiple-steps, poor control over sizes	9–14
Electrochemical carbonization	Size and nanostructure are controllable, stable, one-step	Few small molecule precursors	15–22
Laser ablation	Rapid, effective, surface states tunable	Low QY, poor control over sizes, modification is needed	2 and 23–27
Microware irradiation	Rapid, scalable, cost effective, eco-friendly	Poor control over sizes	28–31
Hydrothermal/solvothermal treatment	Cost effective, eco-friendly, non-toxic	Poor control over sizes	34–40

Figure 2: Synthetic Methods Advantages and Disadvantages [1]

3. Summary and Outlook

This research has described recent progress in the the production and development of CQDs. Specifically, research focuses on synthesis methods, size control, modification strategies, PL properties, luminescent mechanism, and applications in biomedicine, energy conversion and storage, catalysis, and sensor issues.

Although several methods and strategies have been stated for the synthesis of CQDs, creating well-defined structure and sizing is much harder to control. The critical points of synthesis of CQDs is to produce designed structure and size for property studies and selected applications.

While still in the midst of progress, CQDs have already shown enormous potential to play a major role in the production of assays, sensors, bioimaging agents, drug carriers, phototherapy, photocatalysis and electrocatalysis in nanotechnologies. Despite the fact that many optical and electronic properties of CQDs are not yet well known, there is no question that CQDs will play an enormous role on further progress of bioimaging and biomedical research [3]. CQDs are extremely versatile and have the ability to be changed rationally according to specific needs. Applications of CQDs in niche areas such as photocatalysis exemplify the most unforeseen utility of CQDs.

4. Challenges

On the other hand, complicated methods for their isolation, purification, and functionalization, their generally low quantum yields, and complexity in their configuration, design, and structure are some of the problems that need to be resolved before they can fully outperform their quantum dot equivalents for semiconductors.

Quantum dots may also have surface defects that can impact the electrons and holes recombination by serving as temporary "traps." This results in blinking of the quantum dots and deteriorates the quantum yield of the dots [4]. Quantum yield is the amount of absorption versus amount emitted. Once put in living cells, the quantum dots demonstrate accumulation that may interfere with cell function. Often, there is a difficulty in attempting to get quantum dots into cells without destroying the cells in the delivery process.

5. References

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